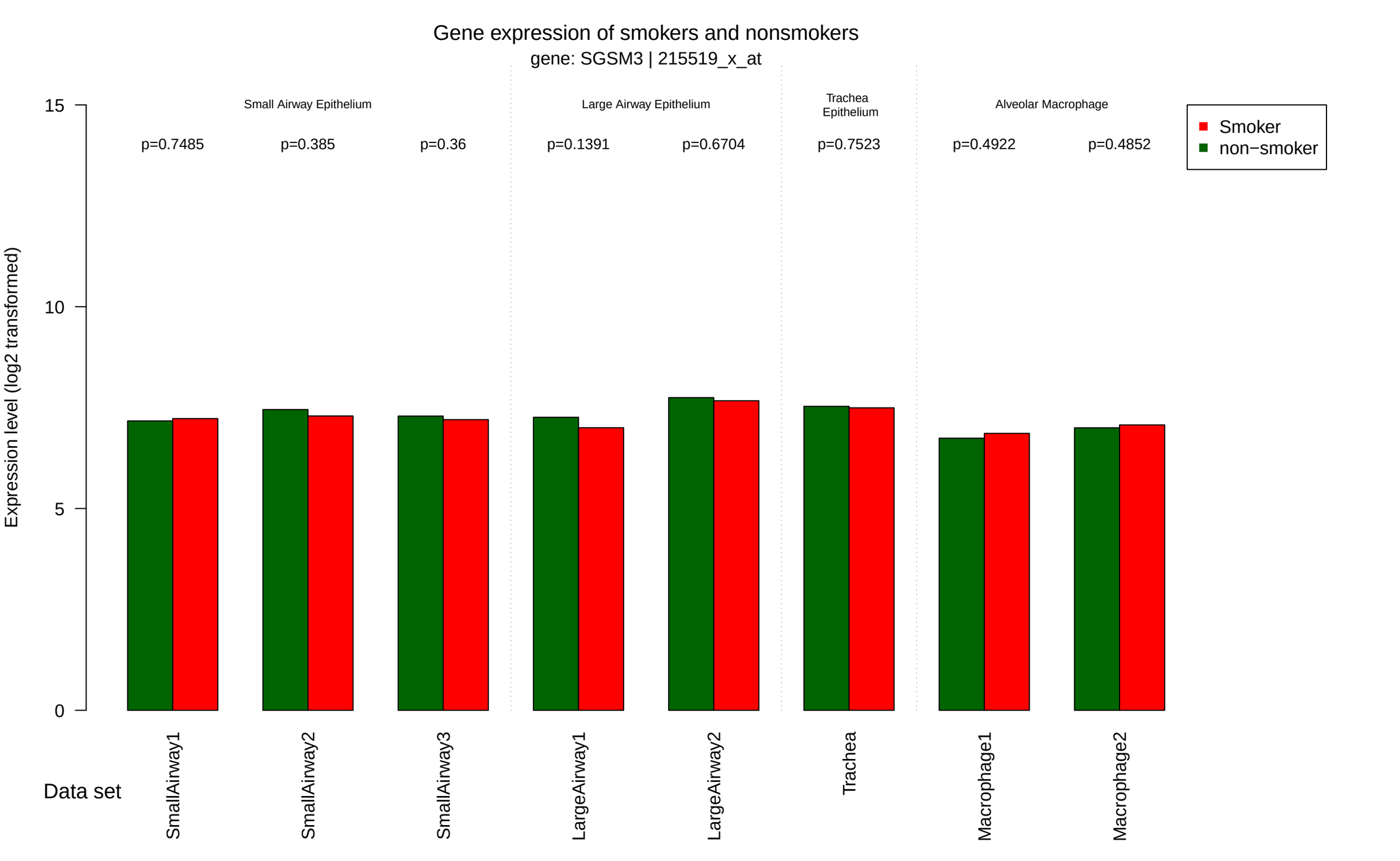
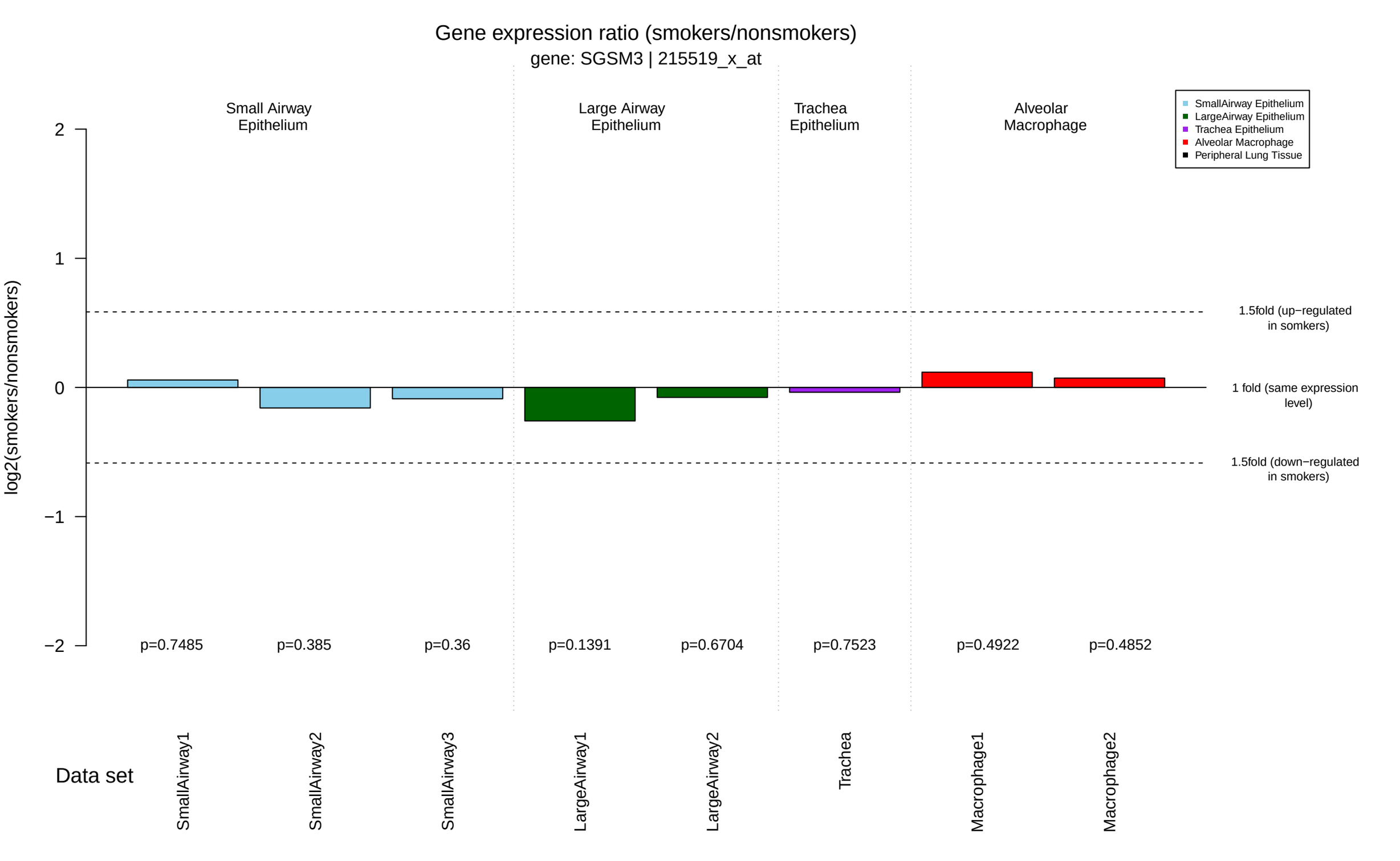
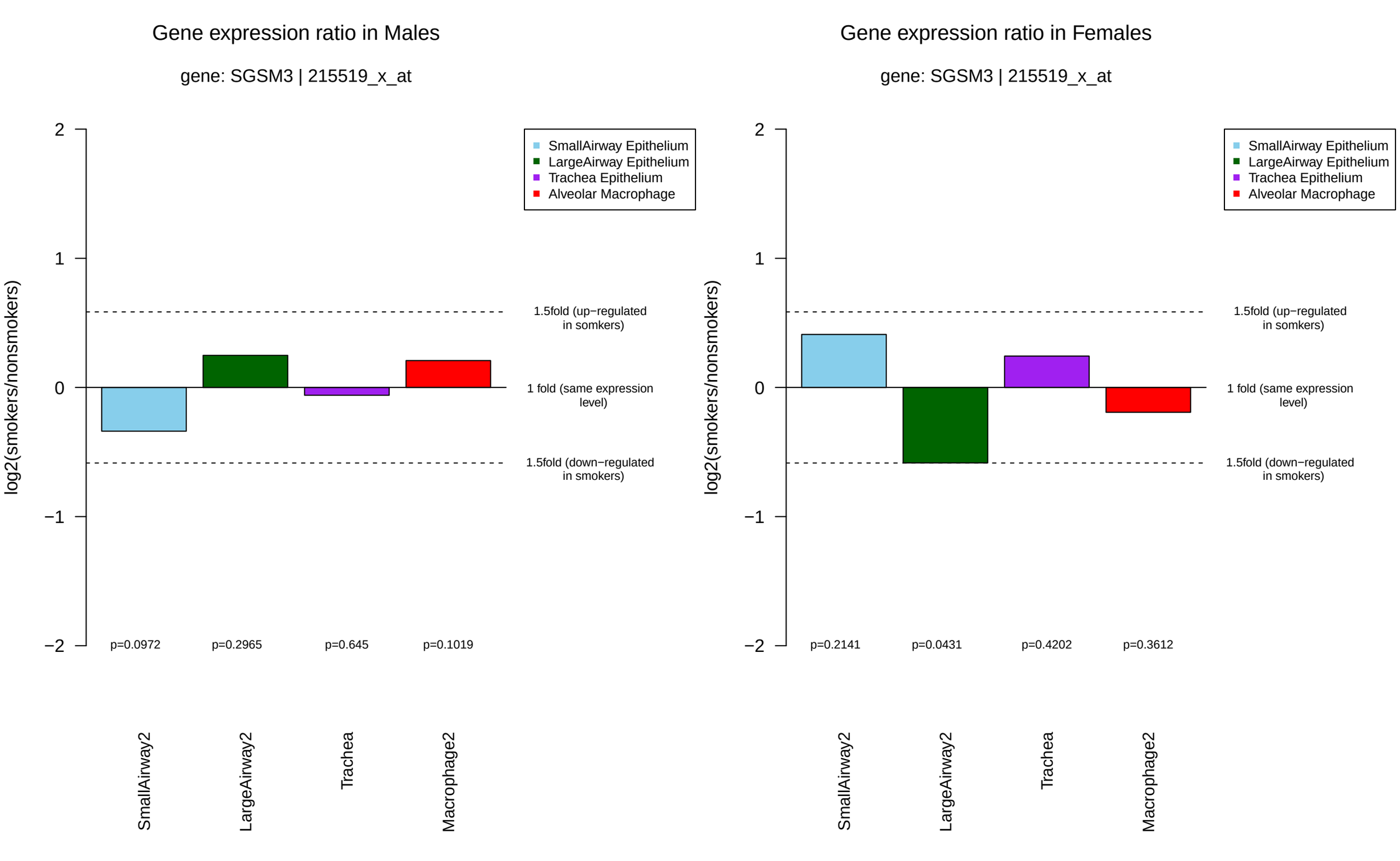
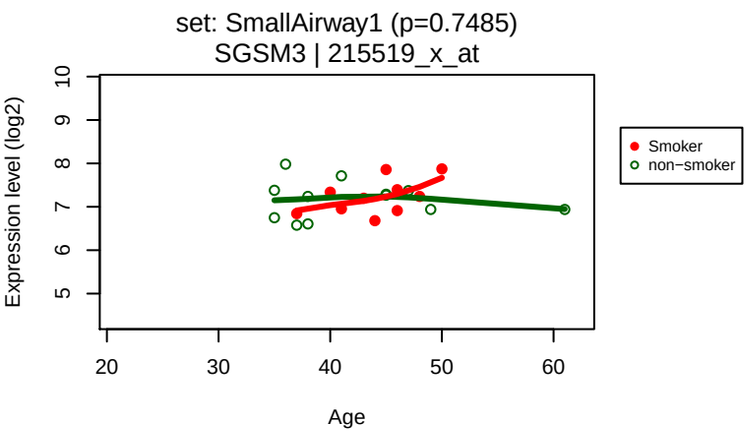
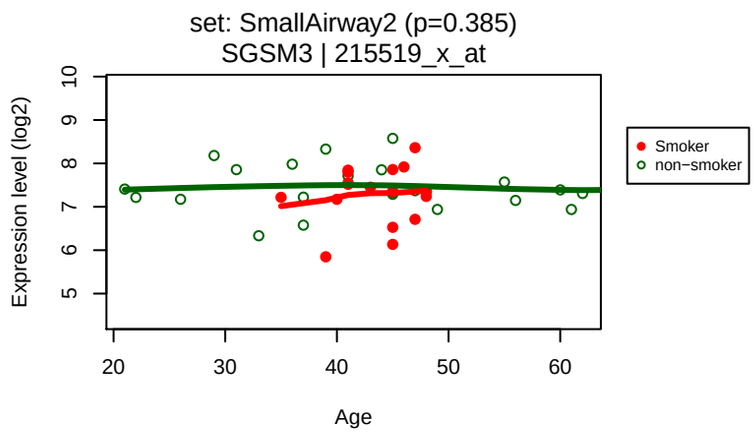
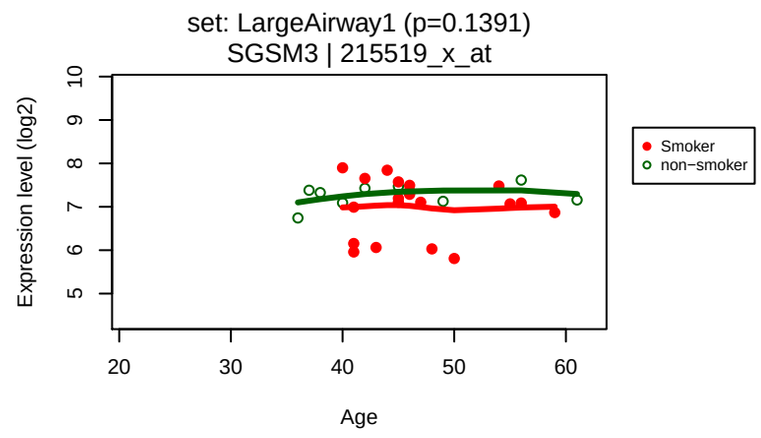

Gene expression of smokers and nonsmokers
gene: SGSM3 | 215519_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7485
p=0.385
p=0.36
p=0.1391
p=0.6704
p=0.7523
p=0.4922
p=0.4852
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

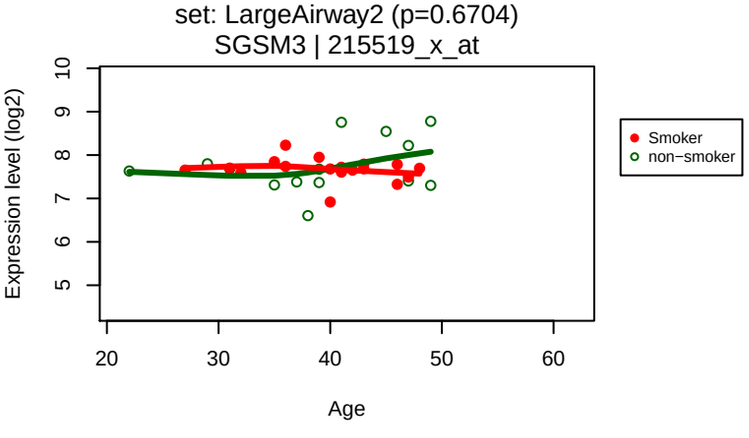

Gene expression ratio (smokers/nonsmokers)
gene: SGSM3 | 215519_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7485
p=0.385
p=0.36
p=0.1391
p=0.6704
p=0.7523
p=0.4922
p=0.4852
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

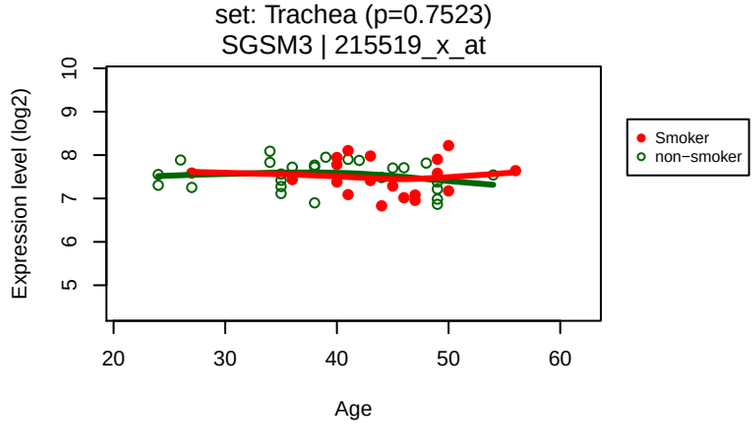

Gene expression ratio in Males
gene: SGSM3 | 215519_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0972
p=0.2965
p=0.645
p=0.1019
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SGSM3 | 215519_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2141
p=0.0431
p=0.4202
p=0.3612
SmallAirway2
LargeAirway2
Trachea
Macrophage2

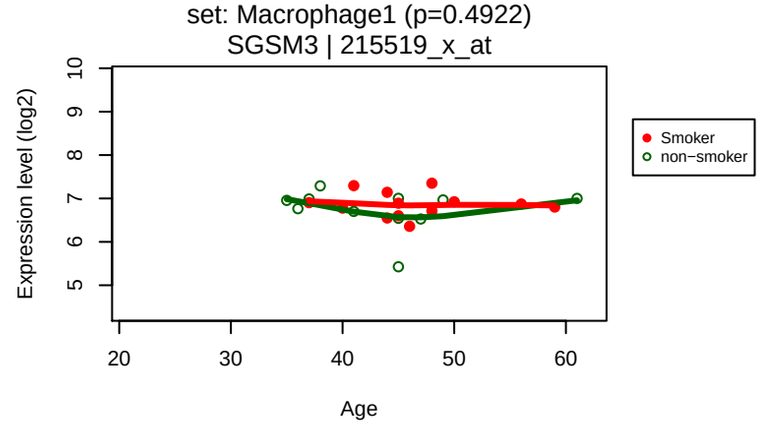

set: SmallAirway1 (p=0.7485)
SGSM3 | 215519_x_at
Expression level (log2)
Age
Smoker
non−smoker

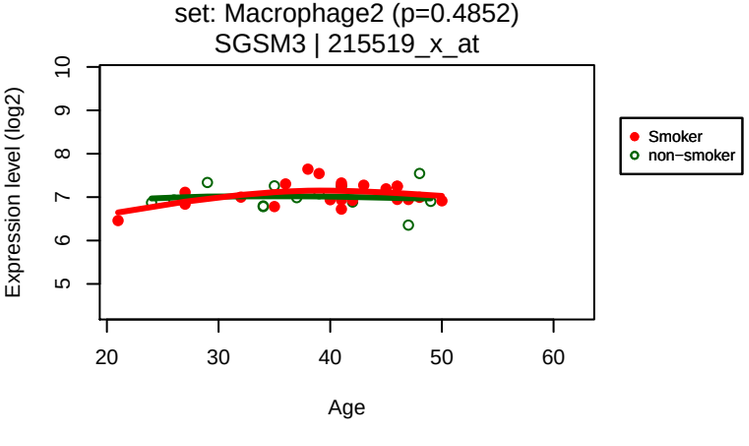

set: SmallAirway2 (p=0.385)
SGSM3 | 215519_x_at
Expression level (log2)
Age
Smoker
non−smoker

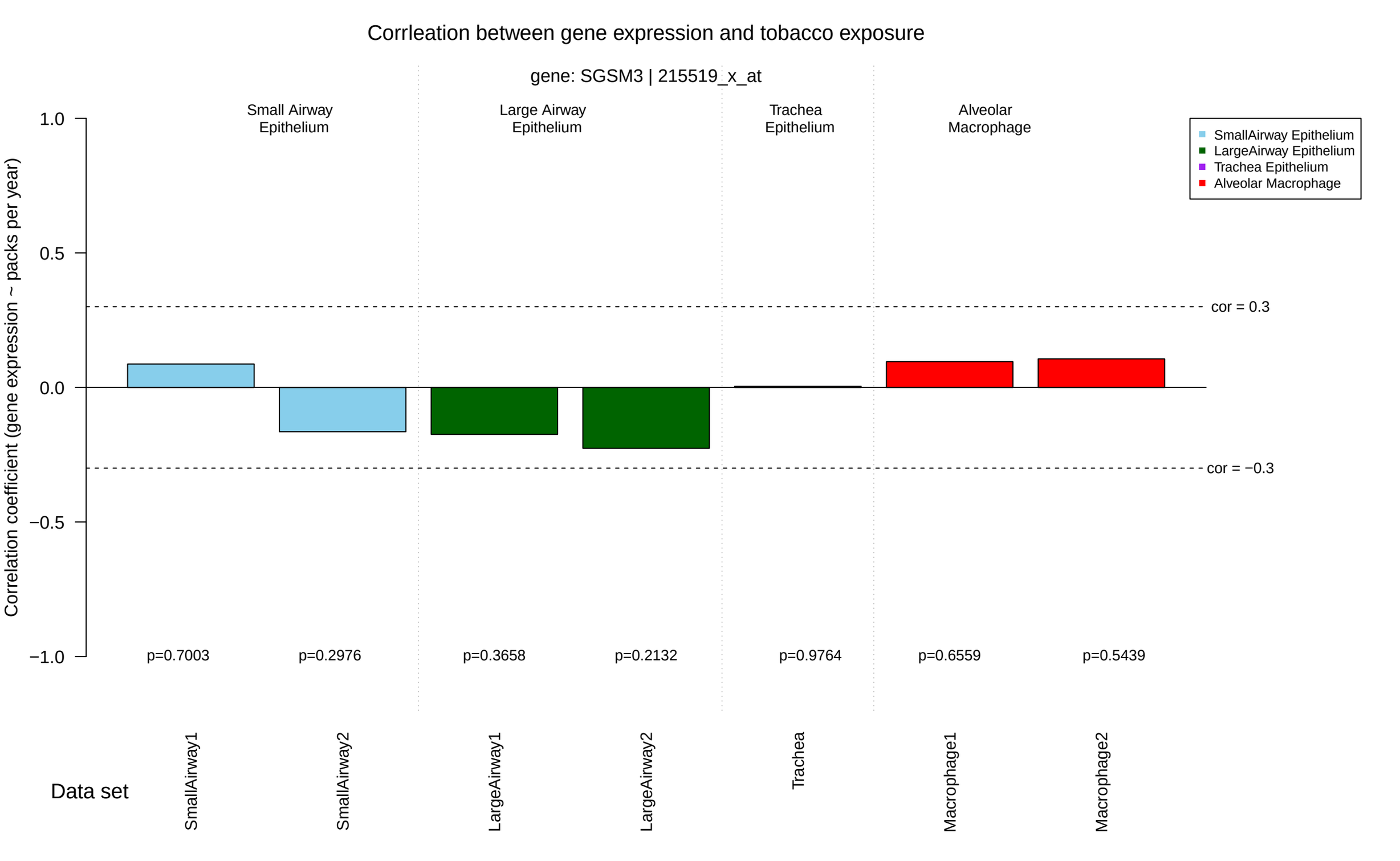

set: LargeAirway1 (p=0.1391)
SGSM3 | 215519_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6704)
SGSM3 | 215519_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7523)
SGSM3 | 215519_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4922)
SGSM3 | 215519_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4852)
SGSM3 | 215519_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SGSM3 | 215519_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7003
p=0.2976
p=0.3658
p=0.2132
p=0.9764
p=0.6559
p=0.5439
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2